

TDDI-Based Charge-Sensing for Aftermarket Defect-Detection in OLED Pixel Circuit with Internal Charging Process

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Abstract

In the aftermarket maintenance and inspection of OLED display panels, it is essential to establish a reliable and cost-effective defect-detection method, especially as pixel circuits may degrade or fail after long-term operation. This paper proposes an internal charging charge-sensing method based on the Touch and Display Driver Integration (TDDI) architecture, which can detect the defect of OLED pixel circuits without external measurement instruments. Compared with conventional external testing methods, the proposed approach offers high circuit integration, low hardware requirements, and direct applicability to both mass-production and aftermarket testing. It effectively shortens the testing process and improves maintenance efficiency, providing a practical and scalable solution for TDDI module's self-diagnostic capabilities of OLED display panels.

Author Keywords

Defect-Detection, OLED Pixel Circuit, Touch & Display Driver Integration, Charge Sensing, Array Testing

1. Introduction

In recent years, OLED (Organic Light-Emitting Diode) displays have become the mainstream technology for portable and automotive panels owing to their self-emissive nature, high contrast, and flexibility. As panel resolution and integration increase, the OLED pixel circuit has grown more complex, typically incorporating multiple thin-film transistors (TFTs), storage capacitors, and compensation modules for stable current and precise gray-level control. While this complexity improves display performance, it also raises the risk of electrical failures, making pixels more prone to open- or short-circuit defects during fabrication or long-term operation. Thus, a reliable and cost-efficient defect-detection method is essential to enhance panel reliability and support aftermarket maintenance.

Conventional defect-detection mainly relies on Automatic Optical Inspection (AOI) and external electrical testing (e.g., Array Test). AOI quickly detects surface anomalies during fabrication but has limited capability for identifying non-visual electrical issues in complex OLED pixel circuits, such as degradation, gate-driver faults, or leakage. External electrical testing offers accurate current-voltage analysis but requires extra probe contact and complex routing, making it impractical once the panel is integrated with the driver IC. In addition, these external methods are time-consuming and costly, unsuitable for large-scale or aftermarket inspection. Hence, developing an electrically integrated diagnostic method capable of identifying defects in complex OLED circuits without external instruments or optical systems is crucial to improving panel reliability and maintenance efficiency.

To overcome these limitations, this paper proposes an internal charging charge-sensing method based on the TDDI architecture for defect-detection in complex OLED pixel circuits. The proposed method utilizes the internal signal source within the OLED pixel

for the charging operation and reads the charge variation at the storage node through the AFE (Analog Front-End) channel in the TDDI. This enables real-time defect identification without any external measurement instruments. Even when the display panel is already integrated into high-value aftermarket products, the proposed method can still perform effective and stable defect-detection, demonstrating strong potential for built-in diagnosis and on-site reliability evaluation in future display systems.

2. TDDI Architecture and the OLED pixel circuit

The TDDI architecture integrates the display driver and touch controller into a single chip, improving system integration, electrical stability, and overall cost efficiency. Within the TDDI framework, the AFE module can be implemented in the form of an integrator as shown in Figure 1, in the pink box. By utilizing the existing data line for voltage measurement, the TDDI can not only drive the pixel array but also switch to a defect-detection mode to perform real-time built-in electrical inspection without relying on external instruments, enabling on-panel defect-detection.

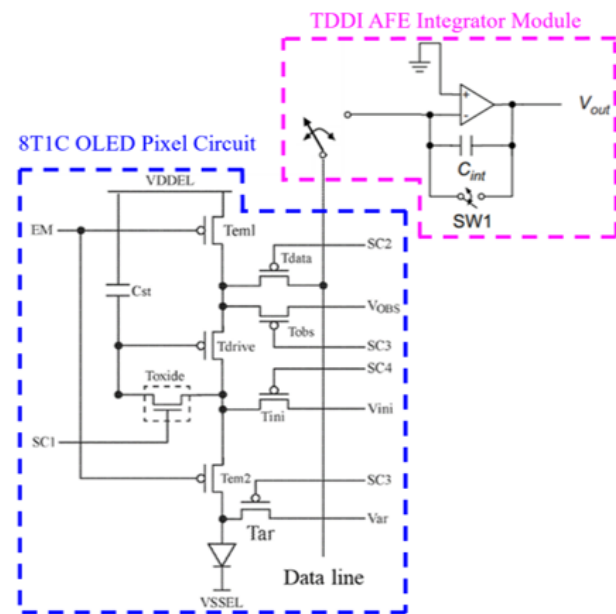


Figure 1. The 8T1C OLED pixel circuit [1] and the TDDI AFE integrator module used for defect-detection.

Figure 1 presents the complex OLED pixel circuit, adopted from an Apple Inc. patent [1], highlighted in the blue box. The circuit consists of multiple TFTs for driving, switching, and compensation, along with a storage capacitor (Cst) that maintains the driving-TFT gate voltage. The pixel operates through initialization, data-writing, and emission phases. While this multi-TFT design improves luminance uniformity, it also increases the

likelihood of open- or short- circuit defects due to process variation and device aging.

In the defect-detection mode proposed in this work, the storage capacitor within the pixel is first charged. During the subsequent discharge phase, the integrator inside the AFE measures the amount of charge released from the capacitor. The presence of pixel defects is then determined based on the variation in the AFE output voltage. A detailed description of the complete defect-detection procedure and signal timing will be presented in the next section.

3. Proposed New Charge-Sensing Method

Compared with conventional charge-sensing defect-detection method [2], which requires switching the data line to the internal test voltage of the source IC and programming the storage capacitor through the data line, this work proposes a new charge-sensing method that eliminates the need for such switching. The key advantage of the proposed method is that the storage capacitor C_{st} can be programmed without using the test voltage inside the source IC, by fully reusing the existing pixel-driving signals such as V_{ini} , V_{obs} , and V_{ar} .

In the newly proposed charge-sensing method, operated under $V_{DDEL} = 5\text{ V}$ and $V_{SSEL} = -2\text{ V}$, V_{ini} first programs -2 V to the storage capacitor C_{st} as shown in Figure 2(a), ensuring proper conduction of T_{drive} during the later write-read out sequence. Next, by turning on T_{obs} , T_{ar} , $T_{em1\&2}$; V_{obs} and V_{ar} continue to program -2 V to C_{st} , as shown in Figure 2(b), ensuring that both the programming and readout operations of the defect-detection path utilize the TFTs and signals within the OLED pixel circuit. After completing the programming step, T_{oxide} is turned off to allow the stored charge to settle, preventing coupling effects [3] that would influence the AFE integrator output if read out immediately. Once the hold period is complete, the charge on C_{st} is read out through the path shown in Figure 2(c) and integrated by the AFE. By observing the resulting integrated output voltage, open- or short-circuit defects can be identified. The complete detection timing diagram and the AFE output voltage under normal conditions are shown in Figure 2(d). Table 1 lists the sizes of TFTs and capacitor values in the OLED pixel circuit in our simulation. The respective SPICE models and device parameters are properly set to fit the typical TFT characteristics.

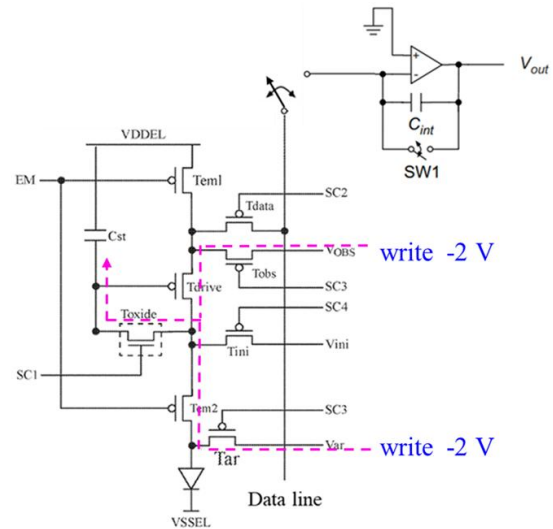


Figure 2(b). After initialization, a voltage is reprogrammed into the storage capacitor via an alternative programming path.

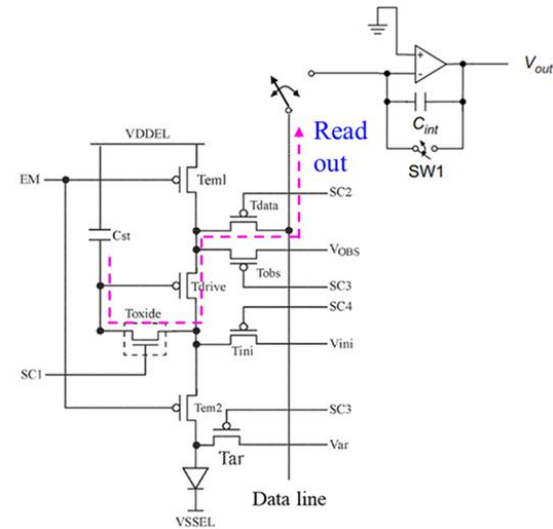


Figure 2(c). The AFE reads out the charge stored in the storage capacitor C_{st} .

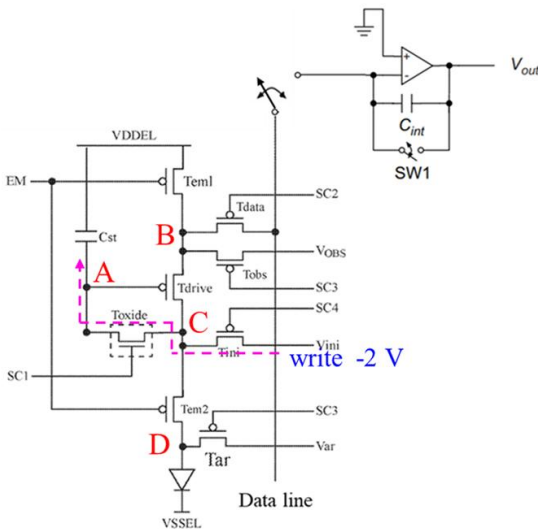


Figure 2(a). Programming the initialization voltage into the storage capacitor C_{st}

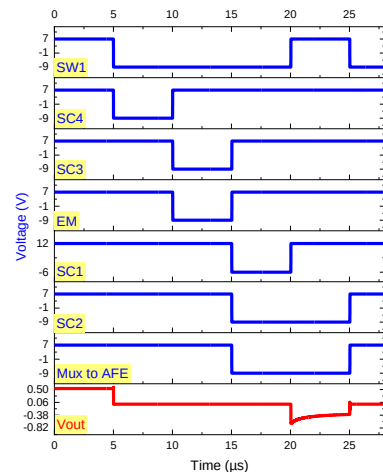


Figure 2(d). Timing Diagram of the Defect-Detection Mode

The newly proposed charge-sensing defect-detection mode offers notable improvements over traditional methods, including reduced switching complexity and power consumption, suppressed parasitic effects, improved measurement stability, and the elimination of additional test-path loading on the panel, resulting in a more reliable and system-integrated defect-detection solution.

Table 1. Sizes of TFTs and Capacitor values in the OLED pixel circuit used in our simulation.

LTPS TFT	W / L
Tem1	20 μm / 2 μm
Tdrive	40 μm / 2 μm
Toxide	20 μm / 2 μm
Tem2	20 μm / 2 μm
Tdata	20 μm / 2 μm
Tobs	20 μm / 2 μm
Tini	20 μm / 2 μm
Tar	20 μm / 2 μm
Capacitor	Capacitance
Cst	0.2 pF
Cint	0.2 pF

4. Defect Case Analysis

Due to the large number of pixel-circuit nodes where open- or short-circuit defects may occur, four representative defect cases are selected to evaluate the new charge-sensing method we proposed in this work. The corresponding AFE output voltage waveforms of each defect case are analyzed and compared with the normal condition to identify characteristic deviations.

When an open-circuit defect occurs between VDDEL and the storage capacitors Cst, as shown in Figure 3(a), Cst loses its ability to charge and discharge and becomes a floating node, causing the voltage at node A to be dominated by parasitic coupling from the data line and TFT switching. This results in abnormal voltage variations during the programming and readout phases. After AFE amplification, this unstable node-A voltage appears as a markedly negative voltage level, providing a clear defect signature as shown in Figure 3(b).

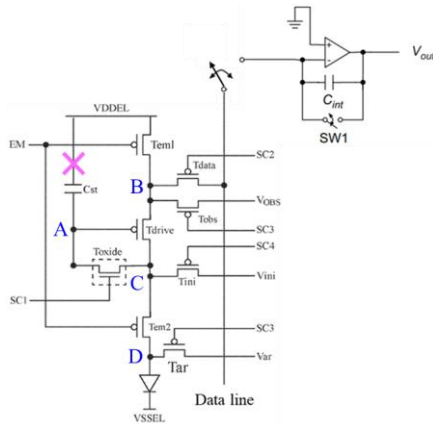


Figure 3(a). Open-circuit defect between Cst and VDDEL.

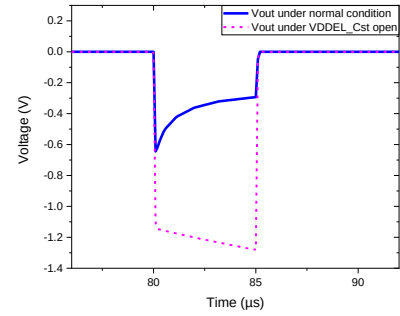


Figure 3(b). The resulting AFE output voltage waveform.

Next, an open-circuit defect between node B and Tem1 is examined, as shown in Figure 4(a), during the programming phase when Tem1 is turned on, the 5 V VDDEL normally pulls node B to a positive voltage. However, when this open-circuit defect occurs, node B is no longer driven by VDDEL and is instead pulled back to the -2 V level provided by Vobs. After this -2 V is programmed into Cst, the AFE output waveform still show the characteristics of capacitor charging and discharging, but its polarity becomes inverted compared with the normal condition, indicating an abnormal output behavior resulting from the incorrect stored voltage as shown in Figure 4(b).

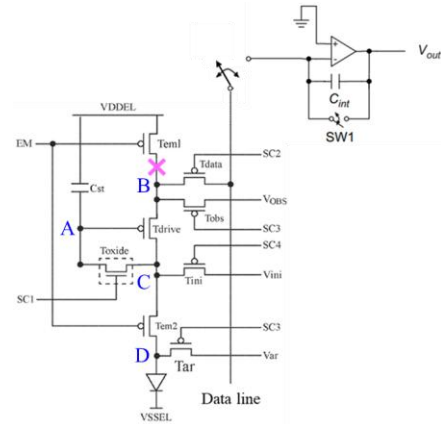


Figure 4(a). Open-circuit defect between node B and Tem1.

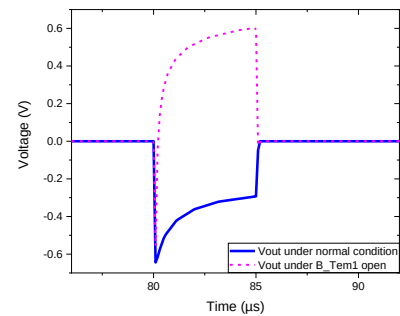


Figure 4(b). The resulting AFE output voltage waveform.

After analyzing the open-circuit defects, we next investigate the short-circuit defect cases within the OLED pixel circuit. As shown in Figure 5(a), when nodes B and C are shorted, the defect effectively forces Tdrive to remain permanently on. During the readout phase, the charge stored in Cst continues to be transferred to the AFE through the B–C short. Even if Tdrive turns off when the voltage at node A falls below its threshold voltage, the stored charge in Cst is still continuously read out by the AFE due to the short between nodes B and C, and this persists until Tdata is turned

off. Consequently, the AFE output voltage becomes significantly higher than in the normal condition, forming a clearly identifiable short-circuit defect signature as shown in Figure 5(b).

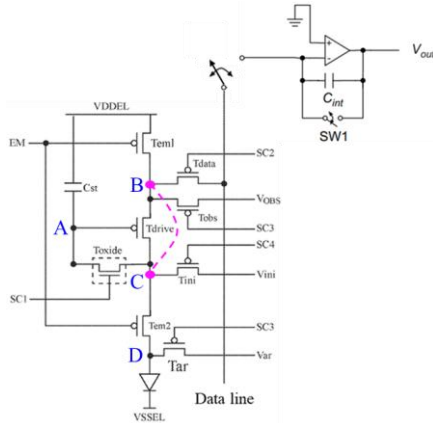


Figure 5(a). Short-circuit defect between nodes B and C.

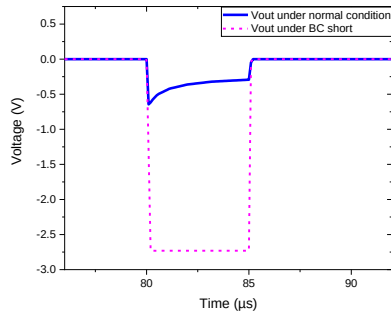


Figure 5(b). The resulting AFE output voltage waveform.

Next, a short-circuit defect between node D and SC3 is examined, as shown in Figure 6(a). When node D becomes shorted to the SC3 signal line, an incorrect voltage is written into Cst. During the programming phase, SC3 normally provides -9 V to turn on Tobs; however, under this short-circuit defect, node D is directly pulled to the -9 V level, causing this unintended voltage to be stored in Cst. As a result, the AFE output shows a clearly different voltage magnitude and polarity compared with the normal condition, forming a distinct signature of this short-circuit defect as shown in Figure 6(b).

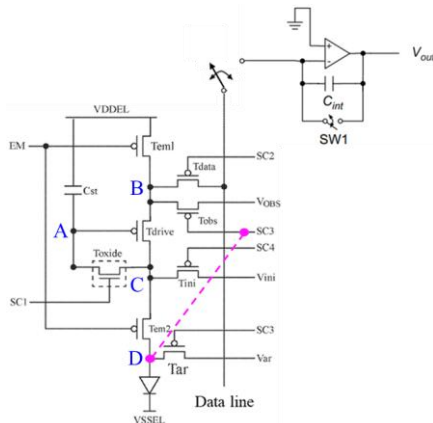


Figure 6(a). Short-circuit defect between SC3 and node D.

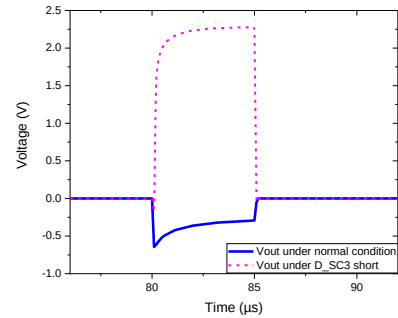


Figure 6(b). The resulting AFE output voltage waveform.

In summary, the analyzed open- and short-circuit defects produce distinct node-voltage behaviors within the pixel, resulting in clearly identifiable waveform differences at the AFE output. Open-circuit defects cause the affected nodes to float, leading to parasitic-coupling-dominated voltage shifts and polarity reversals. In contrast, short-circuit defects disable the intended TFT switching operation or force incorrect biases into Cst, producing abnormal output amplitudes and polarities. These characteristic waveform signatures provide clear diagnostic indicators, demonstrating that the proposed new charge-sensing method can effectively detect defects in complex OLED pixel circuits.

5. Conclusion

In this work, a TDDI-based internal charging charge-sensing method is proposed for detecting open- and short-circuit defects in complex OLED pixel circuits without relying on external measurement instruments. By reusing existing pixel-driving signals and integrating the AFE readout within the TDDI architecture, the method enables fully electrical, on-panel defect inspection suitable for aftermarket maintenance.

The analyzed defect cases demonstrate that different open- and short-circuit faults generate characteristic variations in the stored charge of Cst, resulting in clearly distinguishable AFE output signatures. Open-circuit defects lead to floating pixel nodes dominated by parasitic coupling, while short-circuit defects cause abnormal bias programming or disable proper TFT switching. These distinctive AFE waveform signatures confirm that the proposed method can effectively identify a wide range of pixel defects across complex multi-TFT OLED pixel circuits.

Overall, the internal charging charge-sensing approach provides a reliable, low-cost, and system-integrated solution for built-in electrical diagnosis of OLED display panels, showing strong potential for smart maintenance and on-site reliability evaluation in future TDDI-based display systems.

6. References

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